

Agricultural Vision Failures and ImageNet Transfer in Plants

Agricultural and botanical visual recognition is fundamentally different from general-purpose image classification because the target signal is usually fine-grained, biologically variable, and strongly entangled with context such as crop species, growth stage, lighting, background clutter, and acquisition conditions rather than being a single discrete object category. Plant tasks often require recognizing subtle lesion textures, irregular color changes, mildew patterns, and their spatial arrangement across a leaf or whole plant, while also handling high inter-class similarity, high intra-class variability, and limited labeled data in uncontrolled field environments (Hu et al., 2026; Ranario & Earles, 2025; Picek et al., 2022; Ye et al., 2024).

1. Introduction

Plant vision problems are usually fine-grained recognition problems, not generic object recognition problems. Species, diseases, pests, and growth stages differ by small local cues, soft category boundaries, and organ-specific morphology, while images vary in viewpoint, scale, organ type, posture, and background clutter (Picek et al., 2022; Ye et al., 2024).

Agricultural imagery is also collected under unstructured conditions that are uncommon in benchmark object datasets. Reviews repeatedly identify variable illumination, weather, occlusion by vegetation, diverse plant postures, complex backgrounds, and difficult labeling as central barriers to robust agricultural vision systems (Ghazal et al., 2024; Lei et al., 2024; Lu et al., 2022; Rahman et al., 2024).

Many plant tasks further require joint reasoning over local and global evidence. Disease diagnosis depends on both tiny lesion textures and the distribution of symptoms across the leaf surface, which is why CNNs can do well on localized texture-dominated diseases but often lose robustness when spatial context matters (Hu et al., 2026).

Does agricultural and botanical visual recognition differ fundamentally from general-purpose image classification?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Consensus on whether plant vision is fundamentally distinct

The literature strongly supports that it does differ fundamentally, mainly because agricultural classes are biologically fine-grained and environmentally unstable rather than semantically broad and context-independent.

2. Methods

The search ran over more than 170 million papers in Consensus, spanning Semantic Scholar, PubMed, and other sources. The workflow identified 47,925 papers from the original query, screened 230 with machine-learned relevance filtering, judged 170 eligible after deduplication and expanded citation crawling, and included the top 50 most relevant papers for synthesis.

Search Strategy

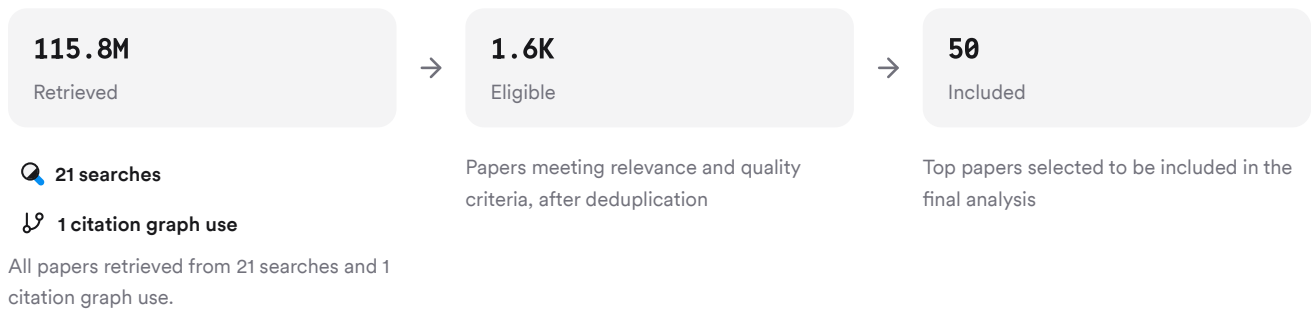


FIGURE 2 Deep-search screening and inclusion workflow

Searches targeted foundational distinctions, domain shift, transfer learning limitations, fine-grained recognition, and evaluation studies across plant disease, species, weed, and crop imagery.

3. Results

3.1 Key Papers

A few papers anchor the corpus because they directly test cross-domain transfer, define pathology-specific recognition demands, or benchmark plant-specific pretraining against ImageNet baselines (Sahili & Awad, 2022; Dong et al., 2023; Hu et al., 2026; Lee et al., 2020).

Paper	Summary
(Wu et al., 2023)	Lab-to-web accuracy collapses from 99.3% to 31.4% (Sahili & Awad, 2022)
(Chen et al., 2020)	ImageNet pretraining gives weak generalization for plant disease tasks (Dong et al., 2023)
(Cui et al., 2023)	Plant disease recognition needs local lesions plus global context (Hu et al., 2026)

FIGURE 3 Key anchor papers in this literature

3.2 Fundamental Differences

Agricultural datasets are usually smaller, less diverse, and less comprehensively annotated than general-purpose datasets, and many are restricted to single crops or limited varieties, which makes overfitting and poor unknown-species adaptation more likely (Lei et al., 2024; Ngugi et al., 2024).

Plant phenotypes also change with environment, development, and organ type, so the same class can look very different over time while different classes can look deceptively similar. This combination of high intra-class variance and low inter-class separation is repeatedly described as the defining difficulty of plant recognition in the wild (Ye et al., 2024; Khalid & Romle, 2023; Seeland et al., 2019).

General-purpose datasets underrepresent crops and often contain image sources unlike agricultural production. ImageNet-scale corpora are large and useful for generic visual priors, but crops occupy only a small fraction of those datasets and the images are often web-crawled rather than collected in realistic field conditions (Zheng et al., 2019; Sahili & Awad, 2022).

3.3 ImageNet Failure Modes

Failure Mode	Evidence	Citations
Domain shift from lab to field	Models trained on controlled images lose validity on field images	(Wu et al., 2023; Salman et al., 2025; Zheng et al., 2019)
Negative transfer from non-botanical priors	High-level ImageNet features can mismatch plant phenotypes	(Sahili & Awad, 2022; Dong et al., 2023)
Background shortcut learning	Segmented or curated training can hurt real-world generalization	(Borhani et al., 2022; Kim et al., 2025)
Crop-domain entanglement	Disease cues couple with crop identity, hurting unseen-crop transfer	(Wu et al., 2026; Lee et al., 2020)
Data scarcity instability	Transfer helps, but general models remain brittle with little plant data	(Sahili & Awad, 2022; Xu et al., 2022; Hu et al., 2026)

FIGURE 4 Main failure modes of ImageNet-pretrained plant models

The clearest empirical failure mode is cross-domain collapse. A GoogleNet model that reached 99.3% on PlantVillage dropped to 31.4% on trusted online images, and PlantVillage-pretrained weights outperformed plain ImageNet weights on PlantDoc-style field imagery, showing that plant-domain proximity matters more than generic visual pretraining alone (Sahili & Awad, 2022).

Another recurrent failure is attention to the wrong evidence. Visualization-based analyses show that transfer-learned models do not necessarily focus on diseased regions, and models trained on segmented or oversimplified inputs can attend to background artifacts or lose contextual richness in new domains (Lee et al., 2020; Kim et al., 2025).

3.4 Mitigations and Alternatives

Plant-specific pretraining consistently appears to outperform plain ImageNet initialization when the target task is botanical. AgriNet models pretrained on 160k agricultural images across 423 classes outperformed ImageNet models on external pest and disease datasets, and plant-disease-specific pretrained backbones were proposed explicitly because ImageNet lacks plant disease domain knowledge (Sahili & Awad, 2022; Dong et al., 2023).

Several studies suggest that architectures with stronger global-context modeling help when disease patterns are spatially distributed. ViTs and CNN-ViT hybrids often outperform standalone CNNs under challenging backgrounds or multispectral conditions, though they typically demand more data or computation and can become unstable in small-sample settings (Thakur et al., 2023; De Silva & Brown, 2023; Shandilya et al., 2025; Hu et al., 2026).

Domain adaptation and disentanglement methods target the lab-to-field and seen-crop-to-unseen-crop gaps directly. Studies report improved cross-domain performance by learning invariant or disease-specific representations rather than relying on generic object features alone (Wu et al., 2023; Wu et al., 2026).

Results Timeline

Plant vision research has moved from lab classification to domain-robust recognition.

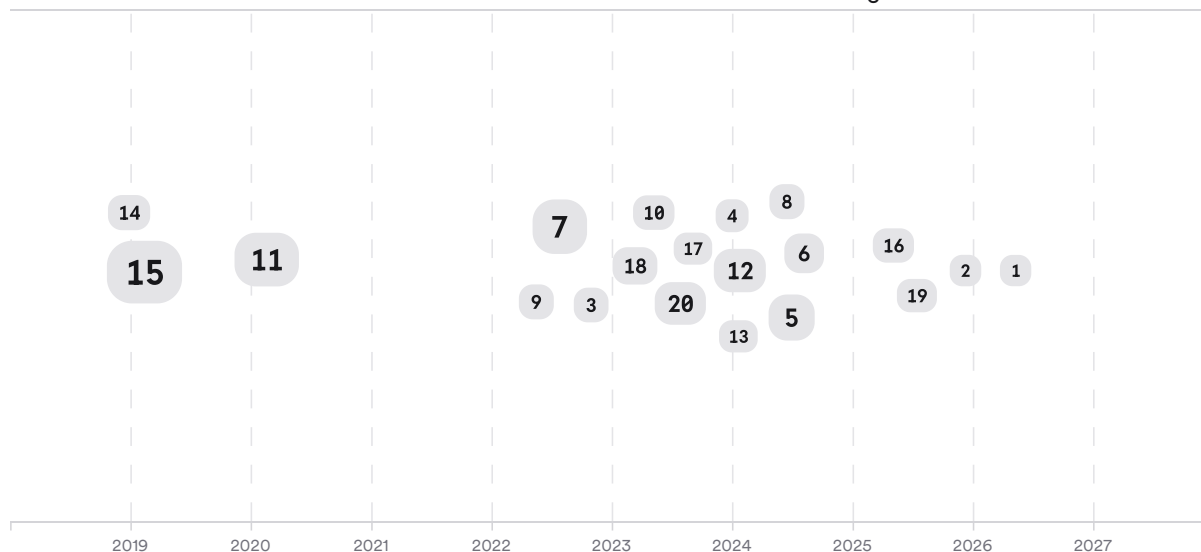


FIGURE 5 Timeline of plant vision studies; larger markers indicate more citations

The timeline shows an early phase dominated by controlled datasets and transfer from ImageNet, followed by critique of lab-to-field generalization, and then a shift toward plant-domain pretraining, transformers, and domain adaptation.

Top Contributors

Based on these 50 papers, transfer learning and plant disease recognition are especially visible in work by **Junde Chen**, **Ali Tufail**, and **Defu Zhang**, while **Frontiers in Plant Science**, **Computers and Electronics in Agriculture**, and **BMC Plant Biology** appear most frequently in this corpus (Chen et al., 2020; Shafik et al., 2024; Shafik et al., 2026).

Type	Name	Papers
Author	Junde Chen	[441e62c2d1a95e5791173f275463dde3][9d8ad8b2830e56b88960837401220e1d]
Author	Ali Tufail	[b1df7ef090b15de4b7df0e5e8d8fca15][034269bdea085c5a9ac2c0d1c5bb4617]
Author	Defu Zhang	[441e62c2d1a95e5791173f275463dde3][9d8ad8b2830e56b88960837401220e1d]
Journal	<i>Frontiers in Plant Science</i>	[0a32900b102a5c66a2750f6e48dea013][340acb7720255843b94e4b97acb73424] [995c49941d59560fae533203a4ac734e][96ce5a616ca15c8e8d3974debc6dd36c] [447d1c5b98cd5a2bb50e7fe4915b865b]
Journal	<i>Computers and Electronics in Agriculture</i>	[dfb7ee9fe3f85ad68894981b245c3000][0051d1d486ee579da0963f6f9dc48359] [9d8ad8b2830e56b88960837401220e1d]

Type	Name	Papers
Journal	<i>BMC Plant Biology</i>	[b1df7ef090b15de4b7df0e5e8d8fca15][034269bdea085c5a9ac2c0d1c5bb4617]

FIGURE 6 Authors and journals appearing most in this corpus

4. Discussion

The corpus is consistent on the main point: high reported accuracies on curated plant datasets do not guarantee real-world robustness. Controlled datasets such as PlantVillage can yield accuracies near 99%, but multiple papers explicitly warn that such results overstate field readiness because the backgrounds are simple, the images are cropped or segmented, and the domain gap to field imagery is large (Sundar et al., 2022; Zheng et al., 2019; Wu et al., 2023).

The critique of ImageNet pretraining is also consistent but more nuanced. ImageNet initialization is useful when data are scarce and often improves convergence and in-domain performance, yet its high-level features do not encode botanical grammar well enough for strong out-of-domain generalization, disease localization, or unseen-crop transfer (Lee et al., 2020; Chen et al., 2020; Sahili & Awad, 2022; Dong et al., 2023).

Recent alternatives point in a common direction: plant-domain pretraining, architectures that combine local and global modeling, and explicit domain adaptation. Evidence for these mitigations is promising across several papers, but it is still less mature than the evidence documenting the failure modes themselves, and many evaluations remain dataset-specific rather than truly standardized across crops, geographies, and imaging conditions (Shafik et al., 2024; Xu et al., 2022; Quan et al., 2023; Richter & Kim, 2025; Hu et al., 2026).

Claim	Evidence Strength	Reasoning	Papers
Agricultural plant vision is a fine-grained, context-sensitive recognition problem	 Strong	Repeated across reviews, benchmarks, and empirical studies	(Hu et al., 2026; Picek et al., 2022; Ranario & Earles, 2025)
ImageNet-pretrained models often lose accuracy under domain shift	 Strong	Direct cross-domain degradation is reported repeatedly and with large effect sizes	(Sahili & Awad, 2022; Wu et al., 2023; Salman et al., 2025)
Plant-specific pretraining improves transfer over generic pretraining	 Moderate	Multiple papers support it, but benchmarks are still fragmented	(Sahili & Awad, 2022; Dong et al., 2023; Quan et al., 2023)
CNN-ViT or hybrid models better capture local plus global disease evidence	 Moderate	Supported by several comparisons, but not yet universal across tasks	(Hu et al., 2026; Shandilya et al., 2025; Thakur et al., 2023)

Claim	Evidence Strength	Reasoning	Papers
Zero-shot general foundation models can replace supervised plant classifiers	 Weak	Current evidence argues against this in agriculture	(Ranario & Earles, 2025)

FIGURE 7 Key claims and evidence strengths in this corpus

5. Conclusion

Agricultural and botanical recognition differs from general-purpose image classification because the relevant signal is subtle, fine-grained, and biologically structured, while the nuisance variation from environment, crop stage, organ, and acquisition is unusually large. The best-supported failure mode of ImageNet-pretrained models is not low benchmark accuracy on curated datasets, but brittle generalization when moving across domains, crops, preprocessing pipelines, and real field conditions.

Research Gaps

The biggest gap is standardized evidence on which pretraining strategy works best across multiple crops, domains, and tasks rather than within a single benchmark.

Topic/Outcome	Pretraining comparison	Cross-domain field tests	Unseen-crop transfer	Interpretability validation
Plant disease classification	9	4	2	3
Plant species recognition	5	6	4	2
Segmentation and localization	4	3	1	2
Zero-shot foundation models	2	2	1	1

Open Research Questions

Future work is converging on models that separate disease semantics from crop identity and preserve both lesion detail and whole-plant context.

Question	Why
Can plant-domain foundation models outperform ImageNet pretraining across many crops and field domains?	Current evidence is promising, but benchmarks remain fragmented and rarely compare domain-specific pretraining at scale.

Question	Why
Which architectural bias best supports lesion texture plus global symptom distribution under limited data?	CNNs, ViTs, and hybrids show different trade-offs, but pathology-oriented comparisons are still sparse.
How can models disentangle disease cues from crop identity to improve unseen-crop generalization?	Cross-crop failure remains a core bottleneck for deployable diagnosis systems in diverse agricultural settings.

Agricultural and botanical visual recognition is therefore fundamentally different from generic image classification, and ImageNet-pretrained models fail most predictably when botanical structure, domain shift, and crop-specific confounding matter most.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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